The Classical Measurement Error (MEC) model

Parametrization

This is an implementation of the classical ME model for a fixed effect. It is best described by an example, let the model be

$$\eta = \beta x + \epsilon$$

where η is the linear predictor, β the effect of the true covariate x with zero mean Gaussian noise ϵ. The issue is that x is not observed directly, but only through w, where

$$w = x + u$$

where u is zero mean Gaussian noise. Even though this setup is possible to implement using basic features ("copy" and multiple likelihoods), we provide the following model which reparametrizes the above,

$$\eta = \nu + \epsilon$$

where $\nu = \beta x$ has the correct distribution depending on various parameters: β has prior $\pi(\beta)$, and x is apriori $\mathcal{N}(\mu_x I, \tau_x I)$[1]. The error is apriori $u \sim \mathcal{N}(0, \tau_u \mathbf{D})$, where τ_u is the observational precision of the error Prec$(u|x)$) with possible heteroscedasticy, encoded in the entries d_i of the diagonal matrix $\mathbf{D}$. The vector s contains the fixed scalings $s = (d_1, \ldots, d_n)$ (with n the number of data points).

Hyperparameters

This model has 4 hyperparameters, $\theta = (\theta_1, \theta_2, \theta_3, \theta_4)$ where θ_2, θ_3 and θ_4 are default set to be fixed (ie defined to be known). The values of θ_2, θ_3 and θ_4 are set to mimic a classical fixed effect, so they will always make sense. To achieve the ME model, please use the appropriate choices for (some of) these parameters!

The hyperparameter specification is as follows:

$$\theta_1 = \beta$$

and the prior is defined on θ_1,

$$\theta_2 = \log(\tau_u)$$

and the prior is defined on θ_2,

$$\theta_3 = \mu_x$$

and the prior is defined on θ_3,

$$\theta_4 = \log(\tau_{\mathrm{x}})$$

and the prior is defined on θ_4.

Specification

The MEC is specified inside the `f()` function as

```
f(w, [<weights>,] model="mec", scale = <s>, values= <w>, hyper = <hyper>)
```

The `w` are the observed values of the true but unknown covariates x, with the *assumption*, that if two or more elements of `w` are *identical*, then they refer to the *same* element in the true covariate x. If data points with identical w values belong to different x values (e.g., different individuals), please add a *tiny* random value to w to make this difference obvious to the model.

The fixed scaling of the observational precision is given in argument `scale`. If the argument `scale` is not given, then s is set to 1.

[1] Note: The second argument in $\mathcal{N}(,)$ is the precision not the variance.

Hyperparameter specification and default values

doc `Classical measurement error model`

hyper

theta1

hyperid `2001`
name `beta`
short.name `b`
prior `gaussian`
param `1 0.001`
initial `1`
fixed `FALSE`
to.theta `function(x) x`
from.theta `function(x) x`

theta2

hyperid `2002`
name `prec.u`
short.name `prec`
prior `loggamma`
param `1 1e-04`
initial `9.21034037197618`
fixed `TRUE`
to.theta `function(x) log(x)`
from.theta `function(x) exp(x)`

theta3

hyperid `2003`
name `mean.x`
short.name `mu.x`
prior `gaussian`
param `0 1e-04`
initial `0`
fixed `TRUE`
to.theta `function(x) x`
from.theta `function(x) x`

theta4

hyperid `2004`
name `prec.x`
short.name `prec.x`
prior `loggamma`
param `1 10000`
initial `-9.21034037197618`
fixed `TRUE`
to.theta `function(x) log(x)`
from.theta `function(x) exp(x)`

constr FALSE

nrow.ncol FALSE

augmented FALSE

aug.factor 1

aug.constr

n.div.by

n.required FALSE

set.default.values FALSE

pdf mec

Example

```
n = 100
beta = 4
prec.y = 1
prec.u = 1
prec.x = 1
## true unobserved covariate
x = rnorm(n, sd = 1/sqrt(prec.x))
## the observed covariate with heteroscedastic scaling
s = runif(n,min=0.5,max=2)
w = x + rnorm(n, sd = 1/sqrt(prec.u*s))
## regression model using the unobserved 'x'
y = 1 + beta*x + rnorm(n, sd = 1/sqrt(prec.y))

## prior parameters
prior.beta = c(0, 0.0001)
prior.prec.u = c(10, 9)
prior.prec.x = c(10, 9)
prior.prec.y = c(10, 9)


formula = y ~ 1 +
f(w, model="mec", scale=s, values=w,
    hyper = list(
        beta = list(
            prior = "gaussian",
            param = prior.beta,
            fixed = FALSE
        ),
        prec.u = list(
            prior = "loggamma",
            param = prior.prec.u,
            initial = log(prec.u),
            fixed = FALSE
        ),
```

```
        prec.x = list(
            prior = "loggamma",
            param = prior.prec.x,
            initial = log(prec.x),
            fixed = FALSE
        ),
        mean.x = list(
            prior = "gaussian",
            initial = 0,
            fixed=TRUE
        )
    )
)

r = inla(formula,
    data = data.frame(y, w, s),
    family = "gaussian",
    control.family = list(
        hyper = list(
            prec = list(param = prior.prec.y,
                initial = log(prec.y),
                fixed=FALSE
            )
        )
    )
)

summary(r)
```

Notes

- `INLA` provides the posteriors of $\nu_i = \beta x_i$ and NOT x_i.
- The posteriors of ν_i come (default) in the order given by the sorted (from low to high) values of `w`. The entry `$ID` gives the mapping.
- The option `scale` defines the scaling in the same order as argument `values`. It is therefore adviced to also give argument `values` when `scale` is used to be sure that they are consistent.